Adaptive Multi-Agent Systems for Scientific Claim Verification

Foundations of AI

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Abstract

Scientific papers make a lot of claims, and not all of them hold up. This project builds a system that takes a scientific claim and a research abstract and decides whether the abstract supports the claim, contradicts it, or simply doesn't say enough to tell. Rather than handing this task to a single model, we split it across multiple specialized agents, each responsible for one part of the problem: finding relevant documents, selecting the most useful sentences, running inference, and deciding how confident to be in the final answer.

We built and compared five different retrieval methods, ranging from classic keyword-based approaches like TF-IDF and BM25 to dense semantic search using sentence embeddings. We also tested what happens when three different NLI models are run on the same claim simultaneously and whether their disagreements tell us something useful about how hard the claim is to verify.

Our main results show that dense retrieval finds the right document more often than any other method, with a Recall@1 of 0.689 compared to 0.605 for TF-IDF. Our best pipeline achieves 49% accuracy in zero-shot mode, while a version fine-tuned directly on SciFact reaches 71.3% accuracy, but only when given clean, pre-labeled evidence. When the fine-tuned model has to rely on retrieved evidence instead, its accuracy drops significantly, which tells us that retrieval quality, not model quality, is the main thing holding the system back. We also found that when three different models disagree on a prediction, that disagreement is itself a useful signal, it reliably identifies the claims that are genuinely ambiguous or hard to verify.

1. Introduction

AI tools are being used more and more to help people navigate scientific literature - summarizing papers, answering questions, checking facts. But there's a problem: these systems are often wrong in ways that aren't obvious. A model might confidently tell you that a paper supports a claim when it actually contradicts it, or it might make up a citation that doesn't exist. In scientific contexts, that kind of overconfidence is genuinely dangerous. This project is about building a system that approaches verification more carefully - one that reads a paper, finds the relevant evidence, and decides whether a given claim is supported, contradicted, or just not addressed.

The task breaks down into three steps. First, you have to find the right paper or abstract from a large collection. Then you have to identify which specific sentences actually relate to the claim. Finally, you have to determine what those sentences say about the claim: do they back it up, push back against it, or neither? Each of these steps is harder than it sounds, and errors compound: if you retrieve the wrong document, nothing downstream can save you.

Prior work has shown that single models struggle with this, especially on harder cases. Evans et al. (2023) tested GPT-3.5 on MSVEC, a benchmark of scientific claims drawn from news sources across ten different fields. The model did okay on supported claims, getting an F1 of 0.635, but dropped to 0.491 on contradicted claims. More troubling was the consistency issue - about one in five identical queries came back with a different answer on a second try. That kind of instability makes a system unreliable for any real use case. Besrour et al. (2025) took a different approach with SQuAI, splitting the task across multiple specialized agents and combining keyword-based and semantic retrieval. That system outperformed single-model baselines by up to 12% across several benchmarks.

We built on both of these. Rather than relying on one model to do everything, we split the work across agents - one for retrieval, one for evidence selection, one for inference, and one for deciding how confident to be in the final answer. We tested five different retrieval methods, including dense semantic search using sentence embeddings, which turned out to work significantly better than keyword-based approaches. We also fine-tuned a model directly on the verification task and found something interesting: it got much better on clean data but actually

got worse in the real pipeline, because it became overconfident and couldn't handle imperfect retrieved evidence. That gap - 71.3% accuracy with clean evidence versus 36-45% in the pipeline - is one of the clearest findings in the project, and it points directly at retrieval as the piece most worth improving.

2. Methods

2.1 Dataset

We use SciFact (Wadden et al., 2020) as our dataset, which contains 1,409 scientific claims paired with abstracts from biomedical research papers. Each claim is labeled as either SUPPORT, CONTRADICT, or NEI (Not Enough Information), and for claims that are supported or contradicted, the dataset also provides the specific sentence indices within the abstract that justify that label - which we use as gold evidence during fine-tuning and evaluation.

After cleaning up duplicate entries, we end up with 866 unique claim-document pairs: 368 labeled SUPPORT, 194 CONTRADICT, and 304 NEI. We split this into an 80/20 train and development set using stratified sampling so that the label distribution stays balanced, giving us 692 training examples and 174 for evaluation.

2.2 System Architecture

The system runs four agents in sequence. The Claim Analyzer categorizes each claim as numeric, comparative, causal, or general - this affects how the Judge Agent scores confidence later. The Retrieval Agent finds relevant documents using one of five methods described in Section 2.3. The Evidence Selector then picks the best sentences across the top three retrieved documents rather than just the top one, which helps when the most relevant document isn't ranked first. The NLI Verifier classifies whether each selected sentence supports, contradicts, or is neutral toward the claim. Finally, the Judge Agent computes a confidence score - if the sentence predictions disagree with each other, confidence drops, and anything below 0.40 gets labeled NEI.

2.3 Retrieval Methods

We built and compared five retrieval methods. TF-IDF and BM25 are both keyword-based approaches, TF-IDF ranks documents by term overlap with the claim, while BM25 adds length normalization and term frequency saturation on top of that. Hybrid retrieval simply combines their scores with equal weighting. Dense retrieval is different, instead of matching keywords, it encodes both the claim and each abstract into vector representations using the all-MiniLM-L6-v2 sentence transformer and retrieves by semantic similarity. This means it can find relevant documents even when the claim and abstract use completely different words to describe the same thing. Finally, Hybrid Dense combines all three methods together, weighting each roughly equally at around 0.33.

2.4 NLI Verification

We use the NLI verifier in two modes. In zero-shot mode, a pretrained DeBERTa model scores each selected evidence sentence against the claim and returns whichever prediction has the highest confidence. No task-specific training is involved, the model is used as-is. In fine-tuned mode, we train the same model directly on SciFact's training set as a three-class classifier, running for five epochs with a learning rate of 1e-5 and gradient accumulation to simulate a larger batch size. We trained on CPU rather than Apple Silicon's MPS backend due to numerical instability issues with DeBERTa on MPS. The checkpoint with the best development F1 is saved and used for evaluation.

2.5 Multi-Agent Verifier

We implement a Multi-Agent Verifier running three independently trained NLI models: DeBERTa (cross-encoder/nli-deberta-v3-small), BART (facebook/bart-large-mnli), and MiniLM (cross-encoder/nli-MiniLM2-L6-H768). These models differ in architecture, parameter count, and training data. Predictions are aggregated by majority voting and confidence-weighted voting. The Judge Agent uses the agreement ratio and standard deviation of agent confidence scores as a disagreement measure, boosting confidence for unanimous predictions and penalizing split votes.

2.6 Evaluation

We evaluate all configurations using overall accuracy, per-class F1, and macro-averaged F1 on the development set. Retrieval quality is measured with Recall@K for K of 1, 5, and 10.

We conduct an ablation study isolating each component's contribution, a single agent versus multi-agent comparison, and a direct evaluation of the fine-tuned model with gold evidence to characterize the retrieval bottleneck.

3. Results

3.1 Retrieval Performance

Table 1 presents Recall@K for all five retrieval methods on the full dataset. Dense retrieval substantially outperforms all other methods at every threshold, achieving Recall@1 of 0.689, Recall@5 of 0.896, and Recall@10 of 0.919. This represents a 8.4 percentage point improvement over TF-IDF at Recall@1 and a 9.9 point improvement over BM25. The Hybrid Dense combination achieves Recall@1 of 0.670, slightly below dense alone, suggesting that adding sparse signals dilutes the dense retrieval advantage at the most critical threshold. BM25 consistently underperforms TF-IDF on this dataset, likely because scientific claims are short queries where term frequency normalization is less beneficial.

These results confirm that semantic similarity captured by sentence embeddings is substantially more effective for scientific claim retrieval than term-based matching. Scientific claims and their supporting abstracts frequently express the same concept using different vocabulary, which dense retrieval handles naturally while sparse methods fail.

Retrieval Method	Recall@1	Recall@5	Recall@10
TF-IDF	0.605	0.841	0.885
BM25	0.590	0.783	0.826
Hybrid (TF-IDF + BM25)	0.632	0.836	0.882
Dense (Sentence Transformer)	0.689	0.896	0.919
Hybrid Dense (All Three)	0.670	0.879	0.919

Table 1. Retrieval Recall@K comparison across five retrieval methods on the full dataset (866 examples).

3.2 Ablation Study

Table 2 presents the ablation study comparing four pipeline configurations using the zero-shot NLI model on 100 development examples. The gold abstract baseline, which bypasses retrieval and uses the correct document directly, achieves 46.0% accuracy and 0.452 macro F1. This represents the theoretical performance for retrieval-based configurations given the zero-shot verifier. TF-IDF retrieval without the Judge Agent matches this accuracy at 46.0% but with a different per-class distribution. Adding the Judge Agent improves accuracy to 49.0% and macro F1 to 0.471, demonstrating that confidence calibration and uncertainty downgrading contributes positively to system performance. The hybrid system achieves 45.0% accuracy with CONTRADICT F1 of 0.242, compared to 0.389 for TF-IDF without the Judge Agent - a drop that reflects the hybrid retriever's different document ranking rather than a degradation from the Judge itself.

The improvement from adding the Judge Agent is the clearest finding from the ablation: calibrated confidence scoring that converts uncertain predictions to NEI reduces false positives for SUPPORT and CONTRADICT, improving overall system reliability even when raw accuracy gains are modest.

Configuration	Accuracy	Macro F1	SUP F1	CON F1	NEI F1
Gold Abstract + NLI (no retrieval)	0.460	0.452	0.506	0.424	0.425
TF-IDF + NLI (no Judge)	0.460	0.442	0.527	0.389	0.411
TF-IDF + NLI + Judge (full system)	0.490	0.471	0.539	0.400	0.474
Hybrid + NLI + Judge	0.450	0.408	0.500	0.242	0.482

Table 2. Ablation study on 100 development examples using zero-shot NLI. Best results in bold.

3.3 Single-Agent vs Multi-Agent Comparison

Table 3 compares single-agent and multi-agent verification configurations using hybrid retrieval on 100 development examples.

The single-agent DeBERTa baseline achieves 43.0% accuracy and a macro F1 of 0.411. Multi-agent majority voting improves raw accuracy slightly to 45.0%, but macro F1 drops to 0.369, and confidence-weighted voting underperforms both, at 41.0% accuracy and 0.365 macro F1. This pattern suggests that naively aggregating multiple models does not reliably improve classification accuracy, consistent with *Evans et al. (2023)*, who observed similar instability with majority voting over GPT-3.5 outputs.

Per-class results, however, reveal an important nuance. Majority voting achieves the highest NEI F1 (0.564) of any configuration, confirming that agent disagreement is correctly channeled into NEI predictions for ambiguous cases.

The most significant finding is qualitative. Consider the claim "Vitamin C reduces the duration of cold symptoms": all three agents disagreed completely, DeBERTa predicted NEI, BART predicted SUPPORT, and MiniLM predicted CONTRADICT. The agreement ratio fell to 33% and the disagreement score rose to 0.1171. Rather than forcing a majority classification, the Judge Agent flagged the output as highly uncertain and downgraded it to NEI with a calibrated confidence of 0.21, well below the 0.40 threshold. This behavior demonstrates that inter-agent disagreement functions as a reliable uncertainty signal, even when individual model accuracy is imperfect.

Configuration	Accuracy	Macro F1	SUP F1	CON F1	NEI F1
Single Agent (DeBERTa)	0.430	0.411	0.500	0.343	0.390
Multi-Agent Majority Vote	0.450	0.369	0.321	0.222	0.564
Multi-Agent Weighted Vote	0.410	0.365	0.521	0.200	0.375

Table 3. Single-agent vs multi-agent verification on 100 development examples with hybrid retrieval.

3.4 Fine-Tuned Model and the Retrieval Bottleneck

Table 4 shows fine-tuning results across five epochs. Dev accuracy climbs steadily from 51.1% at epoch 1 to a peak of 71.3% at epoch 4, before falling back slightly at epoch 5 - a classic sign of overfitting. This peak result substantially exceeds the zero-shot pipeline's best accuracy of 49.0%, confirming that the model learns the task effectively when given clean evidence.

The retrieval bottleneck. When the fine-tuned model is deployed in the full pipeline with retrieved (rather than gold) evidence, accuracy drops to 33 - 45% depending on the retriever - a gap of 25–35 percentage points. This directly quantifies how much performance is lost at the retrieval stage.

The root cause is overconfidence. The fine-tuned model learned to produce high-confidence predictions on clean SciFact-style evidence pairs (mean confidence: 0.87–0.89), but applies that same confidence regardless of retrieval quality. It cannot distinguish well-retrieved evidence from poorly retrieved evidence. The zero-shot model, by contrast, maintains more calibrated uncertainty across a wider range of inputs, which makes it inherently more robust to retrieval errors.

This finding replicates the distribution shift problem directly: the model is trained on one evidence distribution (gold sentences) but evaluated on another (retrieved sentences). The most impactful path forward is pipeline-aware training - training the model on evidence retrieved by the actual retrieval system, so that the training and inference distributions match.

Epoch	Train Loss	Train Acc	Train F1	Dev Acc	Dev F1
1	2.299	0.374	0.334	0.511	0.395
2	0.862	0.624	0.500	0.638	0.517
3	0.557	0.769	0.715	0.690	0.625
4	0.431	0.803	0.769	0.713	0.682
5	0.364	0.825	0.789	0.690	0.619

Table 4. Fine-tuning results across 5 epochs. Dev metrics use gold evidence sentences from SciFact development set.

3.5 Error Analysis

Across 55 total errors on the TF-IDF + NLI + Judge configuration, three distinct patterns emerge: a class difficulty gap, a conservatism bias, and a small but serious overconfidence problem.

Class difficulty - The system handles classes very unevenly. NEI is the easiest at 62.2% accuracy, SUPPORT is moderate at 45.2%, and CONTRADICT is the hardest by a wide margin at just 14.3%. The system rarely succeeds at recognising contradicting evidence.

Conservatism bias - The single most common error - occurring 20 times - is predicting SUPPORT as NEI. CONTRADICT as NEI accounts for another 10 errors. Together, these 30 cases (55% of all errors) share the same root cause: when relevant evidence is not retrieved or selected, the system defaults to uncertainty rather than committing to a verdict. This is appropriate behavior in principle, but its frequency points to retrieval gaps as the primary failure driver.

Claim type difficulty - Comparative claims are the hardest to verify (20.0% accuracy), followed by causal (33.3%), general (44.4%), and numeric (52.6%). The pattern tracks with how much contextual reasoning each claim type requires - comparative claims demand nuanced, multi-part evidence that the retrieval system consistently struggles to surface.

High-confidence errors - The most problematic failure mode: 14 of 55 errors are wrong predictions made with confidence above 0.60. These cases are particularly harmful because the system's stated certainty offers no signal to downstream users or filters that anything has gone wrong.

4. Conclusion

4.1 Summary

We built an adaptive multi-agent system for scientific claim verification, combining five retrieval methods, a Claim Analyzer, single- and multi-agent NLI Verifiers, a fine-tuning pipeline, and a Judge Agent with confidence calibration. Four findings stand out:

- Dense semantic retrieval substantially outperforms sparse methods - Recall@1 of 0.689 vs. 0.605 for TF-IDF.
- The Judge Agent improves zero-shot pipeline accuracy from 46.0% to 49.0% through confidence calibration.
- Fine-tuning DeBERTa on SciFact reaches 71.3% accuracy on gold evidence, exposing a 25–35 point retrieval bottleneck when deployed end-to-end.
- Inter-agent disagreement functions as a reliable uncertainty signal even when it does not improve raw accuracy.

4.2 Strengths

The system is fully modular - each agent can be tested and swapped independently. The Judge Agent's calibrated confidence output is more informative than a bare label, giving downstream users a meaningful signal about prediction reliability. The five-way retrieval comparison offers a thorough characterization of retrieval methods for scientific text, and the finding that dense retrieval substantially outperforms sparse methods is practically useful beyond this dataset. The multi-agent disagreement signal provides meaningful uncertainty estimates at inference time with no additional labeled data required.

4.3 Weaknesses

Three limitations stand out. First, CONTRADICT detection is the system's most persistent failure - pipeline F1 peaks at just 0.424, and only when using gold evidence. Second, the distribution shift between training and deployment is the most consequential gap: the fine-tuned model was trained on clean gold evidence but deployed against noisily retrieved evidence, costing 25–35 percentage points of accuracy end-to-end and largely cancelling out the gains from task-specific training. Third, multi-agent majority voting does not consistently improve over a single agent *(Evans et al., 2023)*, and the 535-document SciFact corpus is small enough that results may not generalize to larger collections where retrieval is harder.

4.4 Future Work

Three directions offer the clearest path forward, roughly in order of expected impact.

Pipeline-aware training is the highest-priority improvement. Rather than training the NLI model on gold evidence sentences, it would be trained on evidence retrieved by the actual retrieval system. This directly eliminates the distribution shift and could push end-to-end accuracy close to the 71% gold-evidence ceiling.

Cross-encoder re-ranking would apply a more powerful model as a second-stage filter over retrieved documents, targeting Recall@1 above 0.80. Better retrieval reduces errors before the verifier sees any evidence, compounding improvements across the rest of the pipeline.

Disagreement-triggered re-retrieval would close the adaptive loop: when agents disagree on a claim, the system would automatically seek additional evidence before committing to a label, rather than passing an uncertain prediction to the Judge Agent.

Works Cited

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